3. ESTIMATION OF GLUCOSE (ORTHOTOLUIDINE METHOD)

Aim

To estimate the amount of glucose present in the given test solution.

Principle

A solution of orthotoluidine in glacial acetic acid when treated with glucose produces a blue coloured product with an absorption maximum at about 640nm. The values obtained represent the true glucose level.

Reagents required

1. Stock Solution

100mg of glucose is weighed and made upto 100ml with distilled water.

Concentration of glucose = 1mg/ml

2. Working standard solution

10ml of stock solution is diluted to 100ml with distilled water.

Concentration of glucose = 100µg/ml.

3. Orthotoluidine reagent

12.5mg of thiourea and 12g of boric acid are dissolved in 50ml of distilled water by heating over a mild flame. 75ml of redistilled orthotoluidine reagent and 375ml of analar acetic acid are mixed separately. The two solutions are mixed and the total volume is made upto 500ml with acetic acid. The reagent is kept overnight at 4˚C.

Procedure

Estimation of glucose

0.5-2.5ml of standard glucose solutions are pipetted out into five different test tubes labelled S1-S5 with the concentration of 50-250µg. 0.5 and 1.0ml of the test solution is pipette out into two different test tubes labelled as T_1 and T_2. Final volume is made upto 2.5 ml using distilled water in all the standard tubes. 4ml of orthotoluidine reagent is added to all the test tubes. A blank is also prepared simultaneously comprising 2.5ml of distilled water and 4ml of orthotoluidine reagent. All the test tubes are heated for 20 minutes in a boiling water bath. The blue colour developed is measured at 640nm using a colorimeter.

A standard graph is drawn with optical density in Y- axis vs. Concentration of glucose in axis. The amount of glucose present in the given test solution is then calculated.

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